

Ahmed Hassan

AI Engineer | Specializing in Edge AI & LLM Systems

ahmedbiniftikhar962@gmail.com | +92 316 1733026

Skills

Languages: Python, C++, Cython, Flutter, React, JS, CSS3, HTML5, MySQL, PostgreSQL

Tech Stack: Artificial Intelligence & Machine Learning (LangGraph, LangChain, LlamaIndex, RAG, LLMs Deployment, Keras, Scikit-learn, NumPy, Pandas) | Deep Learning & Computer Vision (PyTorch, TensorFlow, OpenCV, YOLO, MediaPipe, RetinaFace, DeepFace, DINOv2, RT-DETR) | Backend (FastAPI, Flask, Django, Streamlit UI, REST APIs, API Integration, JWT Authentication) | DevOps (Docker, Git, Linux, CI/CD) | Cloud & Hosting (AWS, GCP, Hostinger, Render, Vercel, Netlify) | Model Deployment (ONNX, TensorRT, Model Quantization)

Professional Experience

Center for Advanced Research in Engineering (CARE), Islamabad

Jul 2025 – Mar 2026

Junior AI Engineer

- Built a hybrid face recognition system leveraging DeepFace (ArcFace), achieving **98%+ verification accuracy** with **real-time processing at 25–30 FPS**, including integrated **anti-spoofing detection**.
- Implemented fallback face matching using MediaPipe FaceMesh (468 landmarks), improving recognition reliability by **~35% in edge cases** where traditional encoding fails.
- Integrated YOLO-based detection with DeepFace pipelines, enabling **multi-face recognition (4–6 faces/frame)** using **8-thread multithreading + queue buffering**, reducing latency by **~40%** and ensuring smooth real-time performance.
- Developed Retrieval-Augmented Generation (RAG) pipelines using LlamaIndex + LLMs, handling **10K+ documents** with **sub-second semantic search latency** and improving response relevance by **~50%**.

BCG X, Boston (Remote)

Feb 2025 - Mar 2025

GenAI Job Simulation

- Performed data extraction and initial analysis on **50K+ records**, implementing data cleaning and preprocessing pipelines that improved data quality by **~40%** and enabled accurate trend discovery and actionable insights.
- Developed an AI-powered financial chatbot using NLP and LLMs, handling **1K+ user queries/day** with **~90% intent accuracy**, delivering real-time financial advice, transaction insights, and personalized assistance.
- Continuously upskilling across AI/ML and full-stack technologies, dedicating **10+ hours/week** to learning, staying aligned with **latest industry trends, tools, and frameworks**.

Deloitte, Sydney (Remote)

Jan 2025 - Feb 2025

AI Data Analytics Job Simulation

- Actively upskilling across AI/ML and emerging technologies, dedicating **10–15 hours/week** to learning, staying aligned with **latest industry trends, tools, and frameworks**.
- Cleaned and analyzed **100K+ data records**, applying statistical and machine learning techniques to uncover patterns, improving decision-making accuracy by **~35%** and generating actionable insights.
- Leveraged advanced digital forensics tools (Autopsy, FTK, Wireshark, Kali Linux) to investigate cyber incidents, successfully analyzing **500GB+ of data**, recovering critical evidence, and supporting **real-world security investigations**.

Education

National University of Modern Languages (NUML), Islamabad

Feb 2023 - Present

Bachelor of Science in Information Technology

CGPA: 3.55/4

Project Works

- **Sleep Detection-Based Automatic System Shutdown (2024):** A real-time system to monitor user alertness using webcam input. Implemented face and eye detection to track eye closure duration and detect drowsiness. Automated system shutdown/lock with a failsafe alert mechanism for user interruption. Python, OpenCV.

- **AutoAI Workflow Builder (2025):** Developed a no-code platform for creating AI-powered workflows using a drag-and-drop interface. Implemented modular blocks for API calls, LLM processing, and automation tasks with real-time execution support. Designed scalable architecture for handling multiple workflows. Python, FastAPI, React.
- **RAG-Powered Document Q&A System (2026):** Built a Retrieval-Augmented Generation (RAG) system for PDF document-based question answering with citation-backed responses. Implemented semantic search using SentenceTransformers and ChromaDB, with a FastAPI backend and Streamlit frontend for real-time interaction. Integrated Gemini (gemini-2.5-flash) to generate accurate, context-grounded answers while minimizing hallucinations.
- **Multi-Agent Research Assistant (2026):** Built an autonomous research system using LangGraph with multiple AI agents for web search, analysis, report generation, and critique-revision cycles. Integrated Groq and Tavily Search for real-time data retrieval and intelligent synthesis, with a Streamlit UI for human-in-the-loop approval and workflow visualization. Generated structured, citation-backed reports with automated PDF export.
- **AI Code Review Agent (2026):** Developed an AI-powered GitHub PR review system using LangGraph and Groq to detect bugs, security vulnerabilities, and code quality issues. Built a FastAPI backend and Streamlit UI to process PR diffs, generate fix suggestions, assign quality scores, and post automated review comments via GitHub webhooks, with secure integration using GitHub tokens, webhook secrets, and ngrok for live endpoint exposure.
- **Pocket-Agent [On-Device LLM Tool Calling System] (2026):** Fine-tuned a $\leq 2B$ parameter LLM using LoRA for structured tool-calling with high argument accuracy and multi-turn reasoning. Implemented model quantization to achieve $\leq 800MB$ size and low-latency ($\leq 200ms$) CPU inference for offline deployment. Built a tool-aware system supporting JSON-based execution (weather, calendar, SQL, etc.), with synthetic data generation, adversarial evaluation, and a reproducible pipeline including training, inference, and demo deployment.

Awards and Certificates

- **Deputy President of NUMCS at NUML:** Organizing and leading peer mentorship initiatives to help students with academic and personal development. Coordinating volunteer opportunities and partnerships with local organizations to enhance community engagement.
- **IBM AI Engineering Professional Certificate**
- **NVIDIA-Certified Associate: Generative AI and LLMs**

Hackathons:

- **Vyrothon** at NSTP by Vyro AI
- **Alstronauts | Space Agents on a mission** by LabLab.ai
- **Green Pioneers** by DevPost
- **Fall in Love with DeepSeek** by LabLab.ai
- **Global Hack Week: AI Major Week 2025** by Major League Hacking
- **Generative AI with IBM Granite** by LabLab.ai

References

References available on Request